Task learning and Classification using Mechanical Neural Networks

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Abstract

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1. Introduction

2. Problem description

This work considers the static analysis of two-dimensional truss structures subjected to large displacements, where the geometric nonlinearity must be taken into account. The truss elements are modeled as linearly elastic members capable of transmitting only axial forces. Nonlinear effects arise solely from the kinematics of deformation.The positional finite element description adopted in this work is based on the formulation proposed by Coda and Greco [1], in which the nodal coordinates are used as primary unknowns instead of displacements.

2.1. Kinematics

Each truss element connects two nodes. The initial (undeformed) positions of the nodes are denoted by (X_1, Y_1) and (X_2, Y_2), while the deformed positions are given by (x_1, y_1) and (x_2, y_2). The reference and current lengths of the element are computed as:

$$L_0 = \sqrt{(X_2 - X_1)^2 + (Y_2 - Y_1)^2} \tag{1}$$

$$L = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2} \tag{2}$$

The axial Green-Lagrange strain is adopted to describe the deformation under large displacements:

$$\varepsilon_G = \frac{L^2 - L_0^2}{2L_0^2} \tag{3}$$

2.2. Constitutive model

The axial force in the truss element is given by a linear elastic relation:

$$N = EA \cdot \varepsilon_G = EA \cdot \frac{L^2 - L_0^2}{2L_0^2} \tag{4}$$

where E is the Young's modulus and A is the cross-sectional area of the element.

2.3. Internal forces

The element's internal force vector, expressed in global coordinates, is defined as:

$$\mathbf{f}_{\text{int}}^{(e)} = \frac{EA \cdot \varepsilon_G}{L_0} \cdot \mathbf{d}^{(e)} \tag{5}$$

with:

$$\mathbf{d}^{(e)} = \begin{bmatrix} x_1 - x_2 \\ y_1 - y_2 \\ x_2 - x_1 \\ y_2 - y_1 \end{bmatrix} \tag{6}$$

This expression accounts for the current configuration of the element, leading to a nonlinear dependence on the nodal coordinates.

2.4. Governing equations

The global equilibrium of the truss is described by the residual vector [2]:

$$\mathbf{R}(\mathbf{d}) = \mathbf{f}_{\text{int}}(\mathbf{d}) - \mathbf{f}_{\text{ext}} = \mathbf{0} \tag{7}$$

where $\mathbf{d}$ is the global vector of current nodal positions, $\mathbf{f}_{\text{int}}(\mathbf{d})$ is the global internal force vector assembled from all elements, and $\mathbf{f}_{\text{ext}}$ is the global external force vector.

3. Design of Mechanical Neural Networks

This section presents the formulation used to optimize the architecture of Mechanical Neural Networks (MNNs). The mechanical structure is represented as a nonlinear truss system, as described in Section 2, and the design variables are the cross-sectional areas of the individual truss elements.

3.1. Optimization problem

The optimization seeks to adjust the cross-sectional areas $\mathbf{A}$ of the elements such that the nodal displacements at selected control nodes match prescribed target displacements. This is achieved by solving a constrained nonlinear optimization problem, in which the equilibrium of the structure must be satisfied, as expressed in Eq. (7).

The objective function is formulated as a least-squares problem, minimizing the discrepancy between actual and desired displacements at the target nodes:

$$\mathcal{J}(\mathbf{A}) = \sum_{i \in \mathcal{N}_{\text{target}}} \left\| \mathbf{d}_i(\mathbf{A}) - \mathbf{d}_i^{\text{desired}} \right\|^2 \tag{8}$$

where $\mathcal{N}_{\text{target}}$ is the set of control nodes, $\mathbf{d}_i$ is the computed displacement of node i, and $\mathbf{d}_i^{\text{desired}}$ is the corresponding desired displacement.

The internal force vector of each element depends on its cross-sectional area, as described in Eq. (5), where the displacement vector $\mathbf{d}^{(e)}$ is defined in Eq. (6).

The optimization problem is formally defined as:

$$\begin{aligned} \text{minimize:} \quad & \mathcal{J}(\mathbf{A}) \quad \text{as in Eq. (8)} \\ \text{subject to:} \quad & A_e^{\min} \leq A_e \leq A_e^{\max} \quad \forall e \\ & \text{with } \mathbf{R}(\mathbf{d}) = \mathbf{0} \quad \text{as in Eq. (7)} \end{aligned} \tag{9}$$

This formulation ensures that, at each step of the optimization, the structure remains in static equilibrium and respects the physical bounds of the cross-sectional areas. The resulting optimized truss behaves as a mechanical analog of a neural network capable of performing specific deformation tasks.

4. Numerical implementation

The numerical implementation of the proposed Mechanical Neural Network (MNN) optimization framework was carried out using the commercial

software COMSOL Multiphysics®. The structural analysis was performed using the nonlinear truss model described in Section 2, with geometric nonlinearity enabled. The design variables, corresponding to the cross-sectional areas of the truss elements, were linked to the structural response through parametric definitions.

For the optimization process, the built-in IPOPT (Interior Point OPTimizer) algorithm available in COMSOL's Optimization Module was employed. IPOPT is a large-scale nonlinear optimization solver well-suited for constrained problems such as the one defined in Eq. (8)–(9).

A flowchart of the optimization procedure is presented in Figure **??**, highlighting the iterative loop between structural analysis and objective function evaluation. Convergence is monitored based on the relative change in the objective function and the norm of the design update vector, with user-defined tolerances set to ensure both numerical stability and design accuracy.

5. Results and discussion

5.1. Task learning through stiffness tuning

A remarkable feature of the Mechanical Neural Network (MNN) is its ability to adapt to different output behaviors under the same input condition by modulating the stiffness distribution. This task learning behavior is achieved by tuning the cross-sectional areas of the truss elements, which act as the structural "weights" of the network.

In this experiment, a single input force is applied to the same initial structure. Two distinct configurations of element stiffnesses are then optimized: one that causes the gripper to close, and another that causes it to open. No change in the actuation is required—only the mechanical architecture is modified.

This illustrates the concept of mechanical task learning: the same mechanical input can produce different functional responses solely by adjusting internal physical parameters. The results are shown in Figure 1, where the two behaviors (closing and opening of the gripper) are clearly observed.

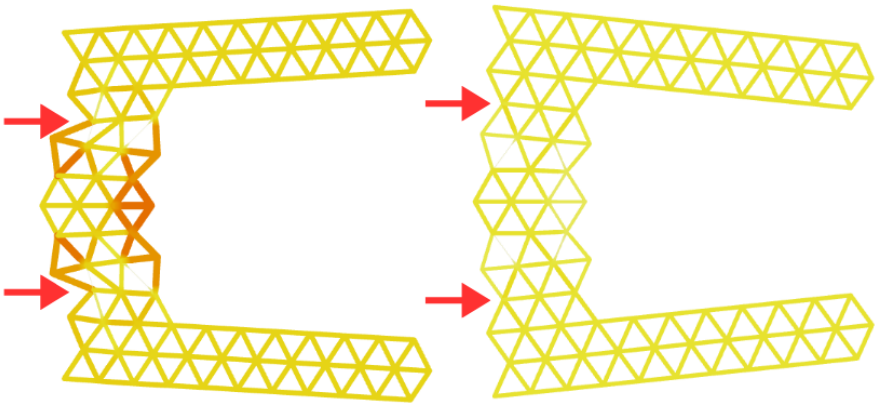


Figure 1: Demonstration of task learning in MNN: with the same actuation input, the gripper either opens (left) or closes (right) depending on the distribution of element stiffnesses.

6. Conclusions

References

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